



VLSI Testing 積體電路測試

Boolean Testing without Fault Model

Professor James Chien-Mo Li 李建模

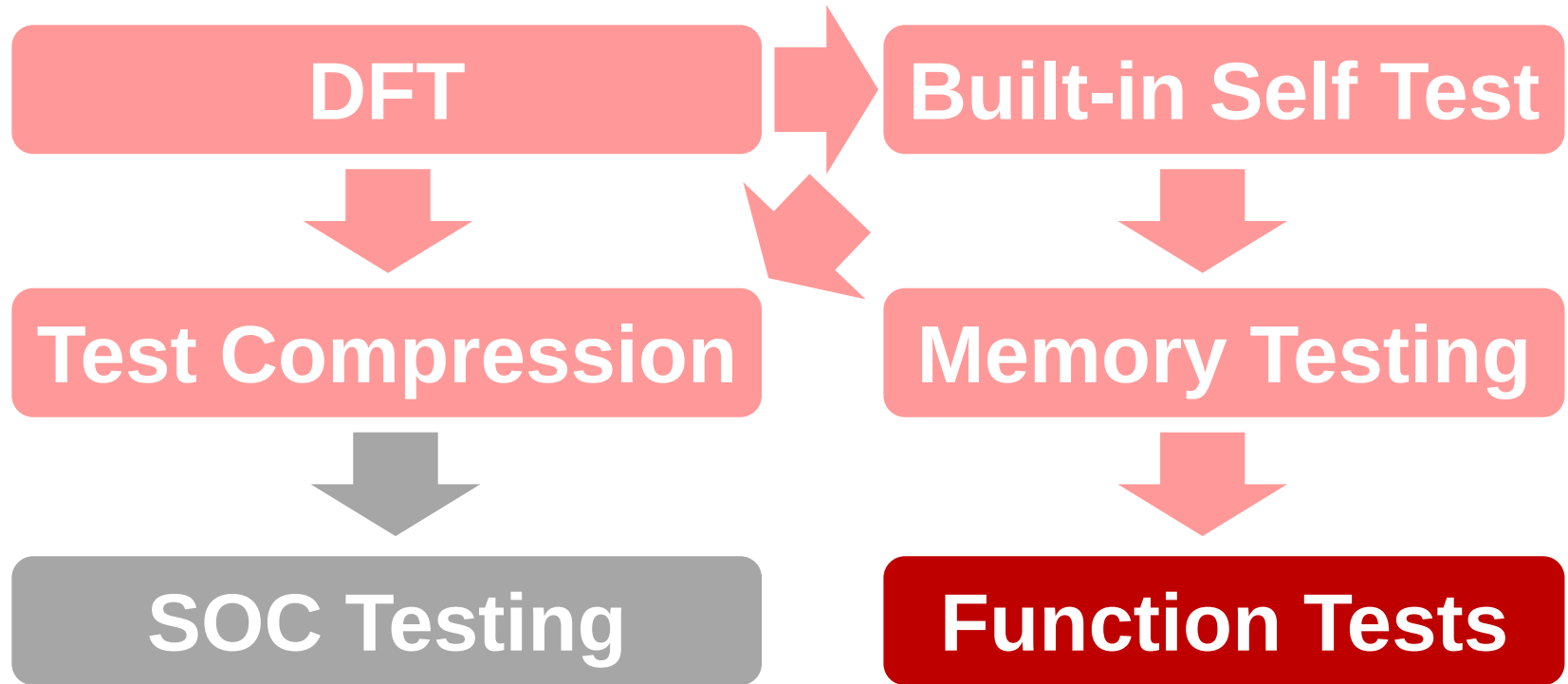
Lab. of Dependable Systems

Graduate Institute of Electronics Engineering

National Taiwan University

(courtesy of Prof. McCluskey, Stanford Univ.)

Course Roadmap (Design Topics)



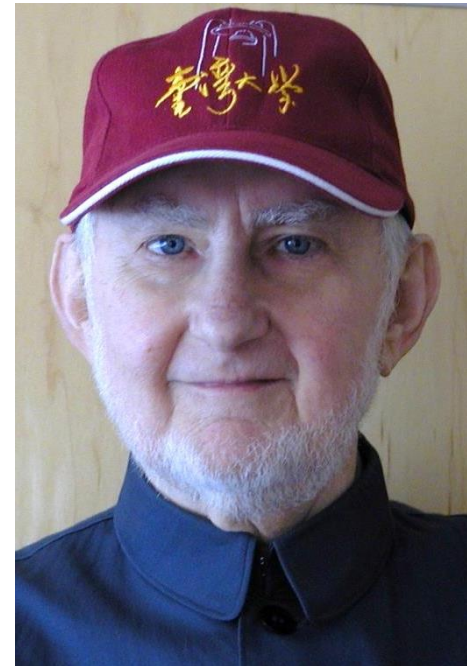
Why Am I Learning This?

- Functional Testing is important because
 - 1) Many circuits still rely on designers to generate test patterns
 - 2) Functional test applied at speed, important for delay defects
 - 3) Helps to debug design errors

*When you see a worthy person, emulate him.
When you see an unworthy person, examine your inner self.
(Confucius) 📖*

Test without Fault Model

- Introduction
- Boolean Tests without Fault Model
 - ◆ Toggle Test
 - ◆ Design Verification
 - ◆ Exhaustive Test
 - ◆ Pseudo Exhaustive Test (PET)
- Conclusions



Many slides in this chapter are in memorial of Prof. McCluskey, CRC, Stanford University

Test Generation

Fault Models	Combinational Circuits (seq. ckt. w/ scan)	Sequential Circuits
No fault model	Toggle Functional Verification Exhaustive Pseudo exhaustive	Checking experiment
Single Stuck-at Fault Model	D PODEN FAN	Extended D 9-valued
Delay Fault Model	Path delay Transition delay	Launch on capture Launch on shift

Testing w/wO Fault Models (review 3.1)

Comparison	Functional testing test ckt functionality w/o fault model	Structural testing test ckt structure with fault model
Test pattern generation	☹ manual	☺ automatic
Fault coverage	☹ low	☺ high
Test speed	☺ at-speed testing test at specified circuit speed helps to defect delay faults	☹ slow speed testing exercise ckt in different ways from functional mode
Test power	☺ low power	☹ high power
Verification / silicon debug	☺ helps to debug	☹ does not help

Two Tests Complement Each Other

Brief History

- **1970~1990**
 - ◆ Silicon expensive, DFT not widely used
 - ◆ ATPG not mature
 - ◆ Design simple
 - ◆ Functional verification tests without fault model popular
- **1990~2010**
 - ◆ Silicon not so expensive, DFT became standard
 - ◆ ATPG was mature
 - ◆ Design very complex, manual test generation infeasible
 - ◆ Structural tests with fault model popular
- **2010~**
 - ◆ Delay defects requires at-speed testing
 - ◆ Test power became serious problem
 - ◆ Functional verification test become popular again

Func. and Structural Tests Both Needed

Test without Fault Model

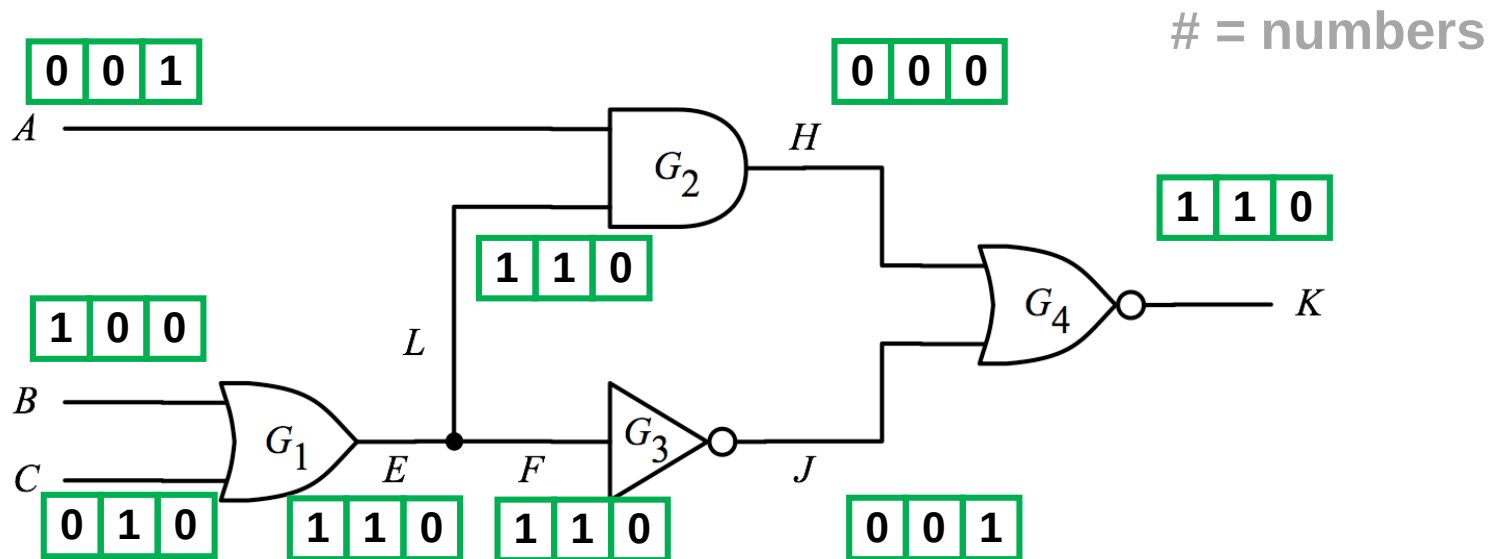
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McCluskey and his collection of hats

Toggle Coverage (DEF-1)

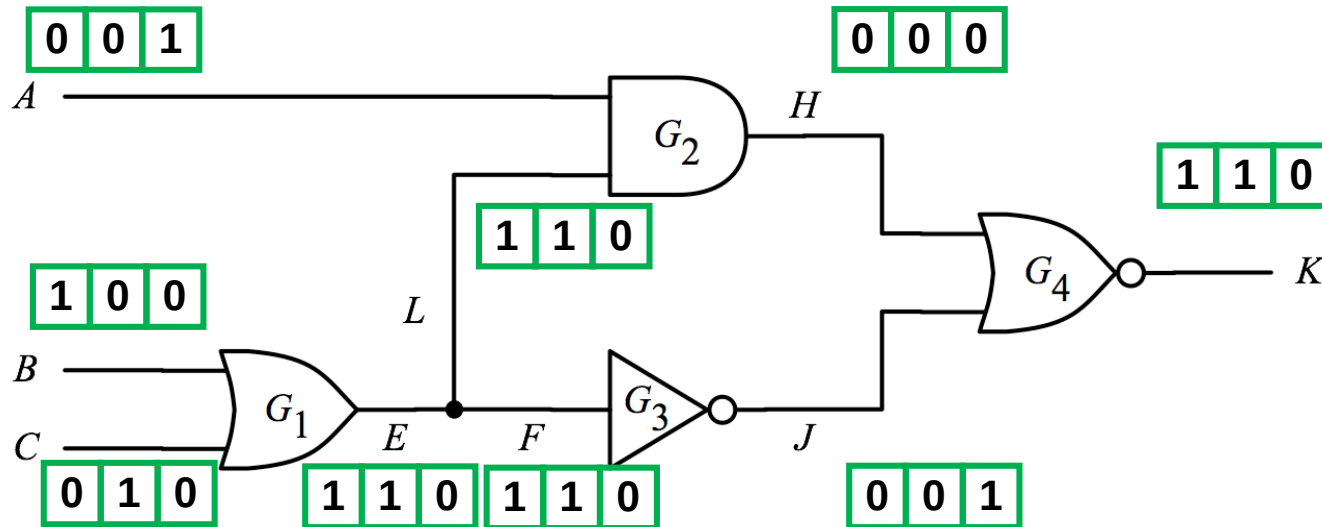
$$\text{toggle coverage} = \frac{\sum_{\text{all nodes } i} \# \text{ of different values of node } i}{2 \times \text{total \# of nodes}}$$



$$\text{Toggle Coverage} = 17/18 = 94\%$$

Toggle Coverage (DEF-2)

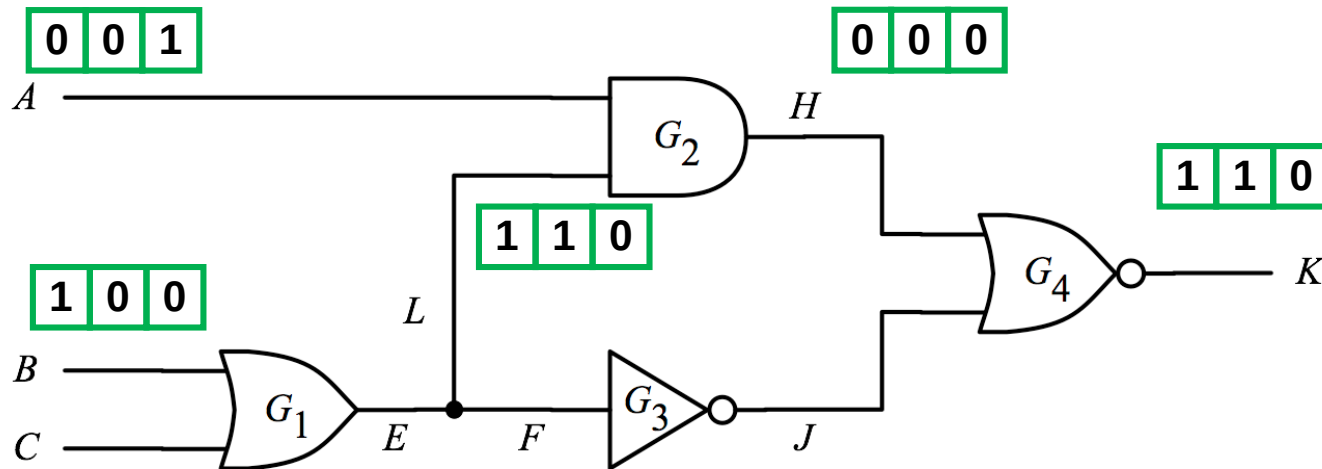
$$\text{toggle coverage} = \frac{\sum_{\text{all nodes } i} \# \text{ of different } \textit{transitions} \text{ of node } i}{2 \times \text{total } \# \text{ of nodes}}$$



Toggle Coverage = 9/18 = 50%
DEF-2 more stringent

Toggle Test: Pros and Cons

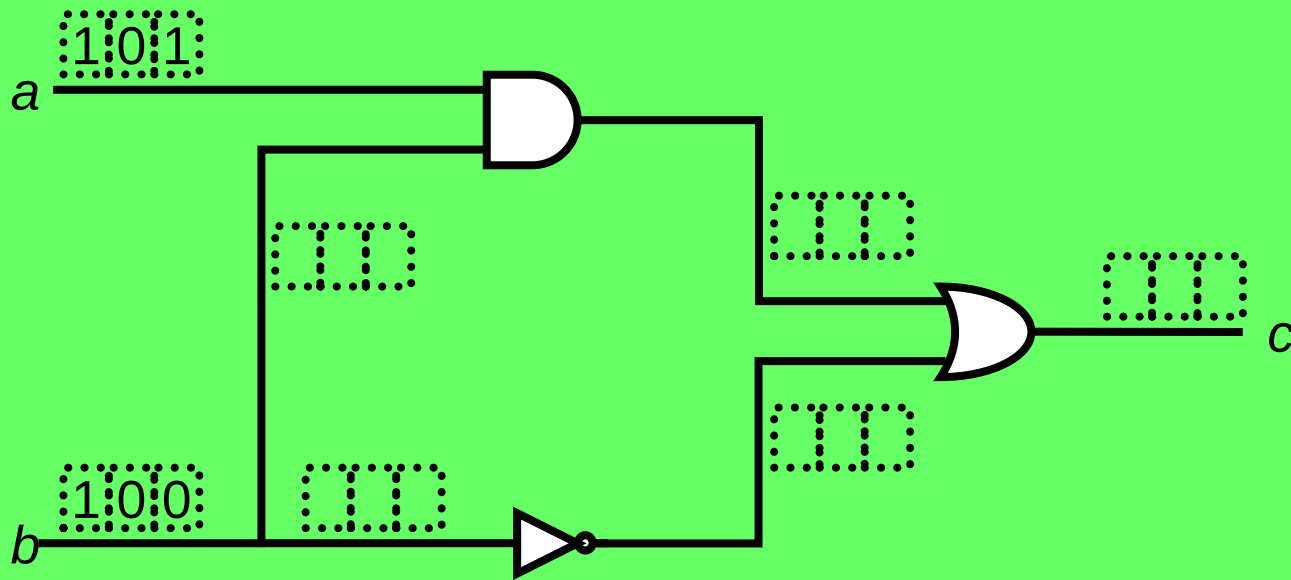
- ☺ Advantage: Toggle coverage is easy to obtain
 - ◆ Logic simulation only, NO fault simulation
 - ◆ Short test length
- ☹ Disadvantage: Toggle coverage is very optimistic
 - ◆ Fault activation only, NO fault propagation



Quiz

Q: Apply 3 patterns to this circuit of 7 nodes.
What is toggle coverage ? Use DEF-2 (transition).

A:



Test without Fault Model

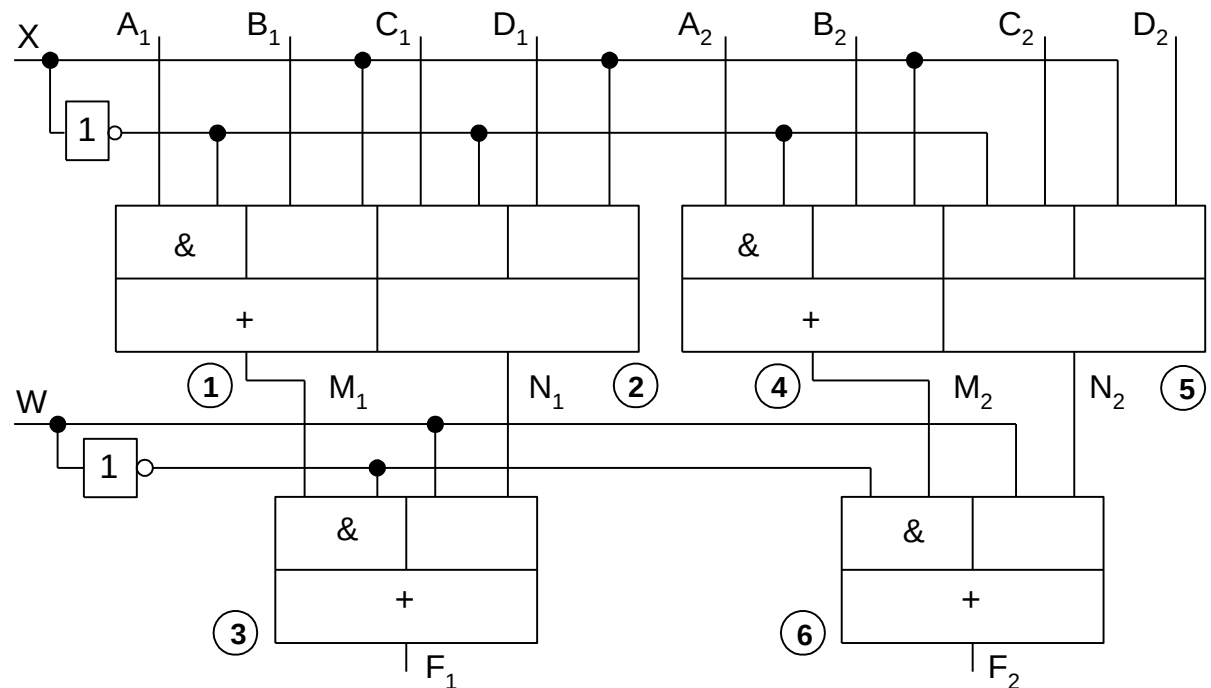
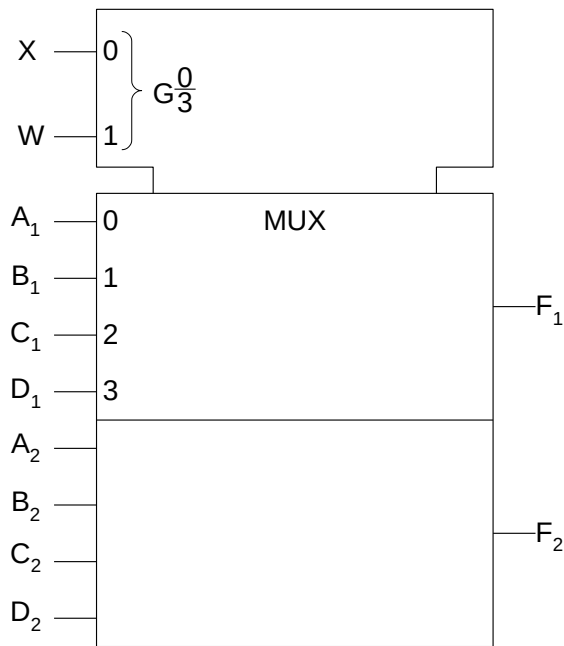
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Design Verification

- **Purpose**
 - ◆ Establish a design correctly implement a behavior specification
- **Created for design verification**
 - ◆ May not good enough for detecting defects
- **Example: Dual 4-to-1 MUX**
 - ◆ $XW=00$ selects $A_1 A_2$, $XW=01$ selects $B_1 B_2$ etc



Low Fault Coverage Problem

- Design verification patterns

- ♦ Test length =8
- ♦ 100% toggle coverage
- ♦ Only 68% SSF coverage

- ATPG test patterns

- ♦ Same test length
- ♦ 100% toggle coverage
- ♦ 100% SSF coverage

W X	A ₁	B ₁	C ₁	D ₁	A ₂	B ₂	C ₂	D ₂	F ₁	F ₂
0 0	0	0	0	0	0	0	0	0	0	0
0 1	0	0	0	0	0	0	0	0	0	0
1 0	0	0	0	0	0	0	0	0	0	0
1 1	0	0	0	0	0	0	0	0	0	0
0 0	1	0	0	0	1	0	0	0	1	1
0 1	0	1	0	0	0	1	0	0	1	1
1 0	0	0	1	0	0	0	1	0	1	1
1 1	0	0	0	1	0	0	0	1	1	1

W X	A ₁	B ₁	C ₁	D ₁	A ₂	B ₂	C ₂	D ₂	F ₁	F ₂
0 0	0	1	1	d	0	1	1	d	0	0
0 1	1	0	d	1	1	0	d	1	0	0
1 0	1	d	0	1	1	d	0	1	0	0
1 1	d	1	1	0	d	1	1	0	0	0
0 0	1	d	d	d	1	d	d	d	1	1
0 1	d	1	d	d	d	1	d	d	1	1
1 0	d	d	1	d	d	d	1	d	1	1
1 1	d	d	d	1	d	d	d	1	1	1

d = don't care

Long Test Length Problem

- Alternative design verification test set 
 - 20 test patterns , 100% SSF coverage

W X	A ₁	B ₁	C ₁	D ₁	A ₂	B ₂	C ₂	D ₂	F ₁	F ₂
0 0	0	0	0	0	0	0	0	0	0	0
0 1	0	0	0	0	0	0	0	0	0	0
1 0	0	0	0	0	0	0	0	0	0	0
1 1	0	0	0	0	0	0	0	0	0	0
0 0	1	0	0	0	1	0	0	0	1	1
0 1	1	0	0	0	1	0	0	0	0	0
1 0	1	0	0	0	1	0	0	0	0	0
1 1	1	0	0	0	1	0	0	0	0	0
0 0	0	1	0	0	0	1	0	0	0	0
0 1	0	1	0	0	0	1	0	0	1	1
1 0	0	1	0	0	0	1	0	0	0	0
1 1	0	1	0	0	0	1	0	0	0	0
0 0	0	0	1	0	0	0	1	0	0	0
0 1	0	0	1	0	0	0	1	0	0	0
1 0	0	0	1	0	0	0	1	0	1	1
1 1	0	0	1	0	0	0	1	0	0	0
0 0	0	0	0	1	0	0	0	1	0	0
0 1	0	0	0	1	0	0	0	1	0	0
1 0	0	0	0	1	0	0	0	1	0	0
1 1	0	0	0	1	0	0	0	1	1	1

Summary

- Functional test without fault models
 - 😊 at-speed testing
 - 😊 low power
 - 😊 helps to debug
 - 😞 manually generated
 - 😞 Low fault coverage
 - 😞 Long test length
- Toggle test
 - ◆ Easy to evaluate
 - ◆ Two definitions: **value, transition**
- Design verification test
 - ◆ Long test length but low FC

